Higher rates of change in electromagnetism

Where do second and higher time derivatives enter: in the material (water's Debye response) or in the field equations (Bopp–Podolsky)? — register entry H19

kronspice simulation report · 2026-11-07 · deterministic

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Summary

The question comes from a premise recorded in *doc/vassilatos-references.md*: that a fast pulse acts on space the way a fast slap acts on water, so that above some rate the medium answers like a solid. In kronspice the answer lives in two places, kept apart. In the **material** (standard physics): a permittivity that depends on how fast the field changes, here water's Debye relaxation, solved in time, in frequency and as a kronspice circuit. In the **field equations** (hypothesis H19): Bopp–Podolsky electrodynamics, which adds a fourth-order term with one new length a, printed beside Maxwell (a = 0) on every line.

- **Water stiffens when hit fast.** A field ramped up in 1 μs sees ε = 78.6; in 10 ps, 36.1; in 1 ps, 9.6 (the high-frequency limit is 5.4). After settling it is always 78.6. The slap-on-water picture is real, and it is Debye relaxation.
- **Three forms of one model agree:** the kronspice AC circuit twin matches ε(ω) from 1 kHz to 100 GHz to 7.9×10^{-13}; the ramp integrator meets the analytic value to 10^{-6}; Kramers–Kronig recovers 73.2 (= ε(0) − ε_∞).
- **Bopp–Podolsky (hypothetical):** a point charge's potential is finite at its centre and falls below Coulomb's by the fraction $e^{-r/a}$, so the effect is gone a few a away.
- **Electron self-energy** becomes finite, $q^2/(8\pi\varepsilon_0 a)$, and equals $m_e c^2$ at $a = 1.409\times10^{-15}$ m.
- **A Coulomb-law test** good to 1 part in 10^6 at 1 cm only bounds a < 0.000724 m; Bhabha scattering bounds it far more tightly (value to be entered from Bufalo et al. 2014). **Status of H19: untested.**

What is being asked

In the material (standard)

Maxwell's constitutive relation D = (1/4π)K E (*Treatise* Art. 619, CLAUDE.md §3c) with a permittivity K that depends on the rate of change of the field. A Debye term carries the first rate of change of the polarization P, $\tau\cdot dP/dt + P = \varepsilon_0\Delta\varepsilon\cdot E$; a Lorentz term the second, $d^2P/dt^2 + \gamma\cdot dP/dt + \omega_0^2 P = \varepsilon_0\Delta\varepsilon\cdot\omega_0^2 E$; and $D = \varepsilon_0\varepsilon_\infty E + \Sigma P_k$ (Debye 1929; Lorentz 1909; Taflove & Hagness ch. 9). In the frequency domain they are $\Delta\varepsilon/(1 + j\omega\tau)$ and $\Delta\varepsilon\cdot\omega_0^2/(\omega_0^2 - \omega^2 + j\gamma\omega)$.

In the field equations (hypothesis H19)

Bopp–Podolsky generalized electrodynamics (F. Bopp 1940; B. Podolsky, *Phys. Rev.* 62, 68, 1942) adds $(a^2/2)\partial_\mu F^{\mu\sigma}\partial^\nu F_{\nu\sigma}$ to Maxwell's Lagrangian, giving the fourth-order field equation $(1 + a^2\Box)\Box A = \mu_0 J$ in Lorenz gauge, with one new length a. **Standard prediction beside every result:** a = 0 — Coulomb's potential, infinite point-charge self-energy, one wave mode.

The model

- **Water at 25 °C** is one Debye term: ε(0) = 78.6, ε_∞ = 5.4 (Elton, arXiv:1704.01667 §II.B), τ = 8.3 ps (Barthel et al. 1990 / Buchner et al. 1999 via Elton Table I). Valid below about 100 GHz: the faster relaxation near 1 ps and the infrared bands are left out.
- **Time domain:** an exponential integrator, exact for a field varying linearly between grid points (*em/dispersion.go*).
- **Circuit twin** (Kron/Branin network analogy, §3a.1): a plate cell of area A and gap d filled with the material is $C_\infty = \varepsilon_0\varepsilon_\infty A/d$ in parallel with one branch per term — a series R–C for Debye ($C_k = \varepsilon_0\Delta\varepsilon A/d$, $R_k = \tau/C_k$) and a series

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R–L–C for Lorentz. For the 1 cm², 1 mm cell: C_∞ = 4.78 pF, C_D = 64.8 pF, R_D = 0.128 Ω. Its admittance is exactly $j\omega(\varepsilon_0 A/d)\cdot\varepsilon(\omega)$, so the netlist solved by kronspice AC is a third form of the same model.

- **Podolsky** (*em/podolsky.go*): static point-charge potential and field, finite self-energy, the two free-wave modes and the Coulomb-test bound.

Results

1. Water at 25 °C (single Debye: ε(0) 78.6, ε∞ 5.4, τ 8.3 ps)

Frequency	ε′	ε″	Loss tangent
1 MHz	78.60	0.00	0.000
100 MHz	78.60	0.38	0.005
1 GHz	78.40	3.81	0.049
10 GHz	62.95	30.01	0.477
19.2 GHz (1/2πτ)	41.95	36.60	0.872
50 GHz	14.79	24.47	1.655
100 GHz	8.00	13.54	1.693

"Slapping" the water with an electric field ramped from 0 to 1 over a rise time T:

Ramp time T	ε seen at the end of the ramp	After it settles
1 µs	78.6	78.6
1 ns	78.0	78.6
100 ps	72.5	78.6
10 ps	36.1	78.6
1 ps	9.6	78.6

Slow, the molecules follow (ε ≈ 78.6). Fast, they cannot turn in time and the water answers like a stiff solid (ε → 5.4). The analytic value at the end of the ramp is $\varepsilon_\infty + \Delta\varepsilon\cdot(1 - (\tau/T)(1 - e^{-T/\tau}))$, which the integrator meets to 10^{-6}.

2. Bopp–Podolsky (hypothetical, H19)

Point charge: $\varphi = q/(4\pi\varepsilon_0 r)\cdot(1 - e^{-r/a})$, finite at the centre $(q/(4\pi\varepsilon_0 a))$. For an electron with a = 1 fm (illustration):

r / a	Maxwell φ [V] (standard)	Podolsky φ [V] (hypothetical)	Fraction below Coulomb
0.0	infinite	1.43996×10^{6}	1
0.1	1.43996×10^{7}	1.37031×10^{6}	0.905
1.0	1.43996×10^{6}	910231	0.368
3.0	479988	456091	0.0498
10.0	143996	143990	4.54×10^{-5}
30.0	47998.8	47998.8	9.36×10^{-14}

- **Self-energy.** Maxwell: infinite. Podolsky: $q^2/(8\pi\varepsilon_0 a)$, equal to $m_e c^2$ when $a = 1.409\times10^{-15}$ m (half the classical electron radius) — Podolsky's motivation.
- **Free waves.** The ordinary Maxwell wave ω = ck, and a massive mode $\omega = c\sqrt{(k^2 + 1/a^2)}$ starting at $\omega = c/a = 3\times10^{23}$ rad/s (4.77×10^{22} Hz) for a = 1 fm. The Podolsky mass is ħ/(ac).
- **Bounds.** A Coulomb-law test good to 1 part in 10^6 at 1 cm bounds only a < r/ln(1/ε) = 0.000724 m. Measured Bhabha (e^+e^-) scattering bounds the Podolsky mass far more tightly (Bufalo, Pimentel & Soto, *Phys. Rev. D* 90, 085012, 2014); that value is to be entered from the paper, not assumed.

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Validation

Comparison	Result
Circuit twin (kronspice AC) against ε(ω), 1 kHz to 100 GHz	worst difference 7.9×10^{-13}
Time-domain ramp against the analytic end value	10^{-6}
Time-domain response against ε(ω) (em/dispersion_test.go)	twin test passes
Causality, Kramers–Kronig: $\varepsilon'(0) - \varepsilon_\infty = (2/\pi)\int\varepsilon''/\omega\, d\omega$	73.2 recovered
Exported water cell, torn automatically into two pieces joined by the Debye branch, against the full solve	2.2×10^{-16}; same power in both (Roth 1959)

Tests: *TestCircuitTwin* (the water cell and a Lorentz test material solved by kronspice AC equal $j\omega(\varepsilon_0 A/d)\varepsilon(\omega)$ to 10^{-9}, §3a rule 4); *TestSlap* (ramp responses equal the analytic end value to 10^{-6} and settle to ε(0)); *em/dispersion_test.go* (water's ε(ω), Kramers–Kronig, time–frequency twin; *TestPodolsky*: potential, field, self-energy and modes); *TestRuns* (the whole report runs).

Limits of the model

- Water is a single Debye term, valid below about 100 GHz; the faster relaxation near 1 ps and the infrared bands are left out, so the 1 ps ramp value is at the edge of the model.
- Podolsky is computed for static point charges and free waves only; a = 1 fm is an illustration, not a fitted value.
- The measured Bhabha-scattering bound on a is not yet entered.

Status

The material part is standard physics and is reproduced three ways. Register entry H19 (Bopp–Podolsky) is **untested**: the standard prediction a = 0 (Maxwell) is printed beside every result, and the effect falls below Coulomb by $e^{-r/a}$, so only very short-range experiments can see it. The Vassilatos slap-on-water analogy is history only, not a physics source.

Reproducing the results

```
go run ./examples/rate_of_change
go test ./examples/rate_of_change ./em
KRONSPICE_EXPORT=1 go test ./examples/rate_of_change -run TestExport   # the water cell's netlist and results sheet
```

Sources: J. C. Maxwell, *Treatise* (1873) Vol. II Art. 619; P. Debye, *Polar Molecules* (1929); H. A. Lorentz, *The Theory of Electrons* (1909); A. Taflove, S. C. Hagness, *Computational Electrodynamics*, 3rd ed. (2005), ch. 9; D. C. Elton, arXiv:1704.01667; J. Barthel et al., *Chem. Phys. Lett.* 165, 369 (1990); R. Buchner et al., *Chem. Phys. Lett.* 306, 57 (1999); F. Bopp, *Ann. Phys.* 430, 345 (1940); B. Podolsky, *Phys. Rev.* 62, 68 (1942); R. Bufalo, B. M. Pimentel, D. E. Soto, *Phys. Rev. D* 90, 085012 (2014); CODATA 2018; G. Vassilatos, *Secrets of Cold War Technology*, p. 27 (history only).

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